I Кинематика и динамика мат. точки.

1.17.

условие $n(t) = \frac{w_n(t)}{g} = \frac{ab^2}{g\sqrt{1+a^2b^2t^2}}$

$v(t) = b\sqrt{1+a^2b^2t^2}$

из 1.15 $\begin{cases} \dot{x} = v\cos\varphi \\ \dot{y} = v\sin\varphi \\ \dot{\varphi} = vk \end{cases}$

$w_n = v^2 k \Rightarrow k = \frac{ng}{v^2}$

$\dot{\varphi} = vk = \frac{ng}{v} = \frac{ab^2 g}{g\sqrt{1+a^2b^2t^2}\, b\sqrt{1+a^2b^2t^2}} = \frac{ab}{1+a^2b^2t^2}$

$\varphi(t) = arctg(abt) + \alpha$

$\cos\varphi = \cos(arctg(abt))$

$\cos\varphi = \frac{\cos\alpha - abt\sin\alpha}{\sqrt{1+a^2b^2t^2}}$

$\sin\varphi = \frac{\sin\alpha + abt\cos\alpha}{\sqrt{1+a^2b^2t^2}}$

$\dot{x} = v\cos\varphi = b\sqrt{1+a^2b^2t^2}\cdot\frac{\cos\alpha - abt\sin\alpha}{\sqrt{1+a^2b^2t^2}} = b(\cos\alpha - abt\sin\alpha)$

$\dot{y} = v\sin\varphi = b\sqrt{1+a^2b^2t^2}\cdot\frac{\sin\alpha + abt\cos\alpha}{\sqrt{1+a^2b^2t^2}} = b(\sin\alpha + abt\cos\alpha)$

$x(t) - x_0 = b\cos\alpha t - \frac{ab^2t^2}{2}\sin\alpha$

$y(t) - y_0 = b\sin\alpha t + \frac{ab^2t^2}{2}\cos\alpha$

повернул СК на α

$\begin{pmatrix} x'-x_0 \\ y'-y_0 \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}\begin{pmatrix} x-x_0 \\ y-y_0 \end{pmatrix}$

$x'-x_0 = b\cos^2\alpha t - \frac{ab^2t^2}{2}\sin\alpha\cos\alpha + b\sin^2\alpha t + \frac{ab^2t^2}{2}\cos\alpha\sin\alpha =$

$= bt$

$y'-y_0 = \frac{ab^2t^2}{2}$

Ответ: $2(y'-y_0) = a(x'-x_0)^2$ в СК: $\begin{pmatrix} x'-x_0 \\ y'-y_0 \end{pmatrix} = \begin{pmatrix} \cos\alpha & \sin\alpha \\ -\sin\alpha & \cos\alpha \end{pmatrix}\begin{pmatrix} x-x_0 \\ y-y_0 \end{pmatrix}$

1.37(g)

$$x=\sigma\tau\cos\varphi$$
$$y=\sigma\tau\sin\varphi$$
$$z=\frac{(\tau^2-\sigma^2)}{2}$$

$$\frac{\partial r}{\partial q_1}=\frac{\partial r}{\partial\sigma}=\begin{pmatrix}\tau\cos\varphi\\ \tau\sin\varphi\\ -\sigma\end{pmatrix} \qquad h_\sigma=\sqrt{\tau^2\cos^2\varphi+\tau^2\sin^2\varphi+\sigma^2}=\sqrt{\tau^2+\sigma^2}$$

$$\frac{\partial r}{\partial q_2}=\frac{\partial r}{\partial\tau}=\begin{pmatrix}\sigma\cos\varphi\\ \sigma\sin\varphi\\ \tau\end{pmatrix} \qquad h_\tau=\sqrt{\sigma^2\cos^2\varphi+\sigma^2\sin^2\varphi+\tau^2}=\sqrt{\tau^2+\sigma^2}$$

$$\frac{\partial r}{\partial q_3}=\frac{\partial r}{\partial\varphi}=\begin{pmatrix}-\sigma\tau\sin\varphi\\ \sigma\tau\cos\varphi\\ 0\end{pmatrix} \qquad h_\varphi=\sigma\tau$$

$$\vec{q}^0_\sigma=\frac{1}{\sqrt{\tau^2+\sigma^2}}\begin{pmatrix}\tau\cos\varphi\\ \tau\sin\varphi\\ -\sigma\end{pmatrix}$$
$$\vec{q}^0_\tau=\frac{1}{\sqrt{\tau^2+\sigma^2}}\begin{pmatrix}\sigma\cos\varphi\\ \sigma\sin\varphi\\ \tau\end{pmatrix}$$
$$\vec{q}^0_\varphi=\begin{pmatrix}-\sin\varphi\\ \cos\varphi\\ 0\end{pmatrix}$$

$$v_\sigma=\sqrt{\tau^2+\sigma^2}\,\dot\sigma$$
$$v_\tau=\sqrt{\tau^2+\sigma^2}\,\dot\tau$$
$$v_\varphi=\sigma\tau\dot\varphi$$

$$\vec v=\begin{pmatrix}\sqrt{\tau^2+\sigma^2}\,\dot\sigma\\ \dot\tau\\ \frac{\sigma\tau\dot\varphi}{\sqrt{\tau^2+\sigma^2}}\end{pmatrix}$$

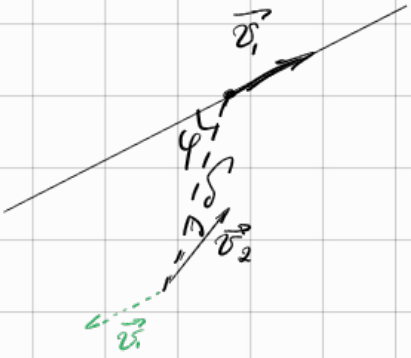
$$v^2=(\tau^2+\sigma^2)(\dot\sigma^2+\dot\tau^2)+\sigma^2\tau^2\dot\varphi^2$$

$$w_\sigma=\frac{1}{h_\sigma}\left(\frac{d}{dt}\left(\frac{\partial(v^2/2)}{\partial\dot\sigma}\right)-\frac{\partial(v^2/2)}{\partial\sigma}\right)=\frac{1}{\sqrt{\tau^2+\sigma^2}}\left(\frac{d}{dt}\left(\dot\sigma(\tau^2+\sigma^2)\right)-\sigma\left(\dot\sigma^2+\dot\tau^2+\tau^2\dot\varphi^2\right)\right)$$

$$w_\tau=\frac{1}{h_\tau}\left(\frac{d}{dt}\left(\frac{\partial(v^2/2)}{\partial\dot\tau}\right)-\frac{\partial(v^2/2)}{\partial\tau}\right)=\frac{1}{\sqrt{\tau^2+\sigma^2}}\left(\frac{d}{dt}\left(\dot\tau(\tau^2+\sigma^2)\right)-\tau\left(\dot\sigma^2+\dot\tau^2+\sigma^2\dot\varphi^2\right)\right)$$

$$w_\varphi=\frac{1}{h_\varphi}\left(\frac{d}{dt}\left(\frac{\partial(v^2/2)}{\partial\dot\varphi}\right)-\frac{\partial(v^2/2)}{\partial\varphi}\right)=\frac{1}{\sigma\tau}\left(\frac{d}{dt}\left(\sigma^2\tau^2\dot\varphi\right)\right)$$

1.34



$$\vec e_r=-v_2\cos\delta+v_1\cos\varphi_1$$
$$\vec e_\varphi=v_2\sin\delta-v_1\sin\delta$$
сидеть



$$\begin{cases} \dot r = \frac{dr}{dt} = v_1\cos\varphi - v_2\cos\delta \\ r\dot\varphi = r\frac{d\varphi}{dt} = v_2\sin\delta - v_1\sin\varphi \end{cases}$$

Ищем $F(r;\varphi)$: $\quad \frac{dF}{dt} = const$

$$\frac{F(v_0;\varphi_0) - F(r=0;\varphi)}{const} = \Delta T$$

$$\frac{dF(r;\varphi)}{dt} = \frac{\partial F}{\partial r}\dot r + \frac{\partial F}{\partial\varphi}\dot\varphi = \frac{\partial F}{\partial r}(v_1\cos\varphi - v_2\cos\delta) + \frac{\partial F}{\partial\varphi}\frac{l}{r}(v_2\sin\delta - v_1\sin\varphi) \Rightarrow$$

$$\boxed{F(r;\varphi) = r f(\varphi)}$$

$$\Rightarrow f(\varphi)(v_1\cos\varphi - v_2\cos\delta) + f'_\varphi(v_2\sin\delta - v_1\sin\varphi) = const$$

$f(\varphi)$ будем искать в виде: $\alpha\cos\varphi + \beta\sin\varphi + \gamma$

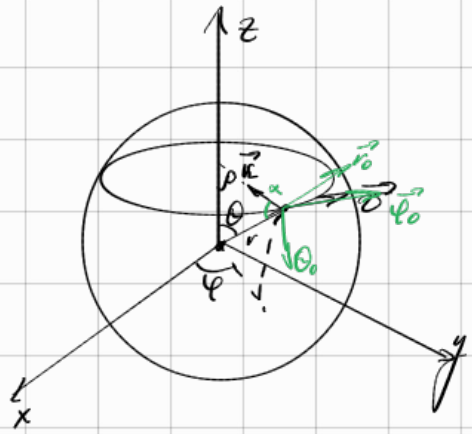
$\alpha = \cos\delta \quad \beta = \sin\delta \quad \gamma = \frac{v_2}{v_1}$

$$T = \frac{r_0\left[(v_2/v_1) + \cos(\varphi_0 + \delta)\right] - r\left[(v_2/v_1) + \cos(\varphi + \delta)\right]}{v_1\left[(v_2/v_1)^2 - 1\right]\cos\delta}$$

1.41

Решение:

$$\begin{cases} x = r\sin\theta\cos\varphi \\ y = r\sin\theta\sin\varphi \\ z = r\cos\theta \end{cases}$$



Найти: $k = \frac{1}{\rho}$

$\vec v \parallel \vec e_\varphi \Rightarrow \vec k = k_r\vec e_r + k_\theta\vec e_\theta$

$\alpha = 90 - \theta$

$k_{r_0} = -k\cos\alpha = -k\sin\theta$

$k_{\theta_0} = -k\sin\alpha = -k\cos\theta$

$\vec k = -k\sin\theta\vec e_r - k\cos\theta\vec e_\theta$

$\rho = r\sin\theta \qquad \vec n = \frac{\vec k}{|k|} = -\sin\theta\vec e_r - \cos\theta\vec e_\theta$

$$\vec k = \frac{\vec n}{\rho} = \frac{-\sin\theta\vec e_r - \cos\theta\vec e_\theta}{r\sin\theta} = -\frac{1}{r}\vec e_r - \frac{ctg\theta}{r}\vec e_\theta$$

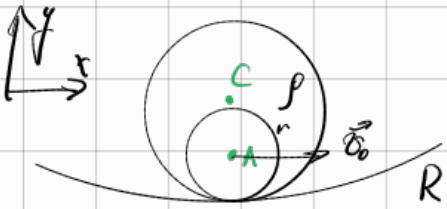
Условие геодезической параметризации: $r = \rho \qquad r = r\sin\theta \Rightarrow \theta = \frac{\pi}{2}$

Плоскопараллельное движение

3.4

Решение:

Найти: ω_{OA}



$v_{OC} = R - \rho$

$r_{CA} = \rho - r$

$r_{OA} = R - r$

$v_0 = \Omega \dot{r}_{OA} = \Omega (R - r)$

$v_C = \Omega \dot{r}_{OC} = \Omega (R - \rho)$

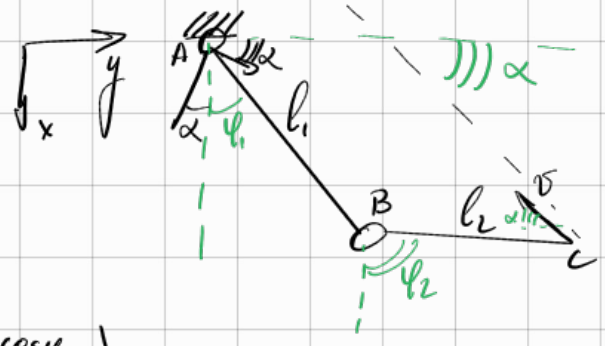
$v_C = \omega \rho \qquad \Omega = \frac{\omega \rho}{R - \rho}$

$v_0 = \omega \rho \frac{R - r}{R - \rho} \Rightarrow \omega = \frac{v_0}{\rho} \frac{R - \rho}{R - r}$

3.39

Найти: $\omega_1(\varphi_1, \varphi_2); \omega_2(\varphi_1, \varphi_2)$

Решение



$\vec{v}_i = \vec{v}_{O'} + \vec{\omega} \times \vec{r}_{O'i}$

$\vec{v}_C = \begin{pmatrix} v_x^B \\ v_y^B \\ 0 \end{pmatrix} + \vec{\omega}_2 \times \begin{pmatrix} l_2 \cos\varphi_2 \\ l_2 \sin\varphi_2 \\ 0 \end{pmatrix}$

$\begin{pmatrix} v_x^B \\ v_y^B \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} + \vec{\omega}_1 \times \begin{pmatrix} l_1 \cos\varphi_1 \\ l_1 \sin\varphi_1 \\ 0 \end{pmatrix}$

$\begin{pmatrix} -v\sin\alpha \\ v\cos\alpha \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \omega_1 \end{pmatrix} \times \begin{pmatrix} l_1 \cos\varphi_1 \\ l_1 \sin\varphi_1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ \omega_2 \end{pmatrix} \times \begin{pmatrix} l_2 \cos\varphi_2 \\ l_2 \sin\varphi_2 \\ 0 \end{pmatrix} = \begin{pmatrix} -\omega_1 l_1 \sin\varphi_1 \\ \omega_1 l_1 \cos\varphi_1 \\ 0 \end{pmatrix} + \begin{pmatrix} -\omega_2 l_2 \sin\varphi_2 \\ \omega_2 l_2 \cos\varphi_2 \\ 0 \end{pmatrix}$

$$\begin{cases} -v\sin\alpha = \omega_1 l_1 \sin\varphi_1 + \omega_2 l_2 \sin\varphi_2 \\ v\cos\alpha = \omega_1 l_1 \cos\varphi_1 + \omega_2 l_2 \cos\varphi_2 \end{cases}$$

$\omega_2 l_2 = -\frac{v\sin\alpha - \omega_1 l_1 \sin\varphi_1}{\sin\varphi_2}$

$v\cos\alpha = \omega_1 l_1 \cos\varphi_1 + \frac{-v\sin\alpha - \omega_1 l_1 \sin\varphi_1}{\sin\varphi_2} \cos\varphi_2$

$v\cos\alpha \sin\varphi_2 = \omega_1 l_1 \cos\varphi_1 \sin\varphi_2 - v\sin\alpha\cos\varphi_2 - \omega_1 l_1 \sin\varphi_1 \cos\varphi_2$

$v(\sin\alpha\cos\varphi_2 + \sin\varphi_2\cos\alpha) = \omega_1 l_1 (\sin\varphi_1 \cos\varphi_2 + \sin\varphi_2 \cos\varphi_1)$

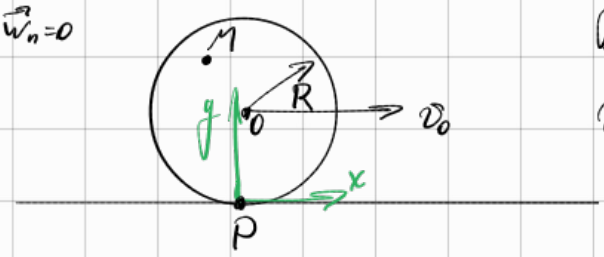
$v\sin(\alpha + \varphi_2) = \omega_1 l_1 \sin(\varphi_1 - \varphi_2) \Rightarrow \omega_1 = \frac{v\sin(\alpha+\varphi_2)}{l_1 \sin\varphi_2 - \varphi_1}$

аналогично $\omega_2 = -\frac{v\sin(\alpha+\varphi_1)}{l_2 \sin\varphi_2 - \varphi_1}$

3.26

Решение

Найти: ТМТ: $\vec{w}_n = 0$



Введем СК в ней удобнее будет.

$v_0 = \omega R$

$\vec{r}_{PM} = x\vec{i} + y\vec{j}$

$\vec{v}_m = \vec{\omega} \times \vec{r} = \begin{pmatrix} 0 \\ 0 \\ -\omega \end{pmatrix} \times \begin{pmatrix} x \\ y \\ 0 \end{pmatrix} = \begin{pmatrix} \omega y \\ -\omega x \\ 0 \end{pmatrix}$

$\vec{w}_m = \vec{w}_P + \vec{\varepsilon} \times \vec{r} - \omega^2 \vec{r}_{PM}$ $\quad \omega = const \Rightarrow \vec{\varepsilon} = 0$

$\vec{w}_P = \vec{w}_O + \vec{\varepsilon} \times \vec{r} - \omega^2 \vec{r}_{PO}$ $\quad \vec{v}_O = 0$

$\vec{w}_P = \begin{pmatrix} \omega^2 R \\ 0 \end{pmatrix}$

$\vec{w}_m = \begin{pmatrix} \omega^2 R \\ 0 \end{pmatrix} - \omega^2 \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -\omega^2 x \\ \omega^2 (R - y) \\ 0 \end{pmatrix}$

Условие $\vec{w}_n = 0$: $\vec{w}_m \parallel \vec{v}_m$

$\vec{w}_m \times \vec{v}_m = 0$ $\quad \begin{pmatrix} -\omega^2 x \\ \omega^2 (R-y) \\ 0 \end{pmatrix} \times \begin{pmatrix} \omega y \\ -\omega x \\ 0 \end{pmatrix} = 0$

$\omega^3 x^2 - \omega^3 (R - y) y = 0$

$\omega^3 (x^2 - yR + y^2) = 0$

$x^2 + \left(y - \frac{R}{2}\right)^2 = \left(\frac{R}{2}\right)^2$

